NurseSync AI

Intelligent Audio-Based Clinical Handoff System

Project Proposal for UCS503P

Submitted to: Dr. Asif

Thapar Institute of Engineering and Technology

Shivansh Sharma(Roll No: 1024030732)

Ishaan Bhalla(Roll No: 1024030950)

Pushp Batheja(Roll No: 1024030941)

August 15, 2026

1 Higher-order goal

This project aims to improve patient safety and continuity of care by reducing the time, inconsistency, and error rate involved in nurse-to-nurse clinical handoffs. While the underlying goal — safer, more reliable communication of patient information — is broader than any single application, the immediate engineering objective is to translate structured, accurate handoff documentation into a measurable, deployable software solution: a system that lets a nurse speak a handoff and converts that speech into structured, searchable, and auditable clinical records.

2 Time-to-value

- We will deliver meaningful increments quickly by building the core patient/handoff workflow (login, patient list, patient profile, structured handoff storage) before adding any AI/ML capability.
- We will prioritize fast feedback over perfection by following a staged roadmap — development environment, backend, database, patient system, mobile frontend, audio capture, speech-to-text, NLP extraction, summary/timeline, RAG assistant, ML risk model, testing, deployment — validating each stage before moving to the next.
- Success in the first iteration is defined narrowly: a nurse can log in, view a patient, and create and retrieve a structured (even if manually entered) handoff record. AI/ML layers are added incrementally on top of this working core.

3 Problem Statement

Nurses routinely hand off patient information verbally or in free-text notes at the end of a shift — for example, describing a patient's diagnosis, vitals, medications administered, and pending tests in a single spoken narrative. Reports indicate nurses can spend a substantial share of a shift on manual charting, time taken directly from patient care. Common pain points include:

- **Fragmentation:** handoff information is typed manually, spread across multiple notes, or communicated purely verbally, causing details to be lost or inconsistently recorded.
- **Inconsistent structure:** different nurses document differently, making handoffs harder to scan quickly during a busy shift change.
- **Poor searchability:** once written, information is difficult to search or retrieve later (e.g., ”what was this patient's oxygen level this morning?”).
- **Handoff errors:** incomplete or misinterpreted information during nurse-to-nurse handoffs is a recognized contributor to preventable medical errors and missed critical details.
- **Accountability gaps:** without reliable, time-stamped records, it is difficult to reconstruct exactly what happened and when during a patient's care.

Why this matters now (contextual relevance): On a busy ward, even small improvements in how quickly and accurately information is captured and transferred can meaningfully reduce missed or delayed care.

4 Proposed Solution

4.1 Overview

Build a mobile-first ”speak-to-structured-handoff” system with the following components:

- **Voice handoff capture:** a nurse records a spoken handoff for a specific patient instead of typing it.
- **Speech-to-text:** the recorded audio is transcribed to a text transcript.
- **Medical entity extraction (NLP):** the transcript is parsed into structured fields — diagnosis, medications, vitals, allergies, tests, and events.
- **Structured SBAR handoff + AI summary:** extracted information is organized into a standard Situation/Background/Assessment/Recommendation format and a short, scannable summary.

- **Timeline generator:** every extracted event (medication given, vitals recorded, test ordered) becomes a timestamped entry in a longitudinal patient timeline.
- **Missing-information detection:** the system checks a handoff against an expected field set (diagnosis, vitals, allergies, pending tests, recommendation) and flags gaps.
- **RAG-grounded AI assistant:** nurses can ask natural-language questions about a patient ("what changed today?") and receive answers grounded strictly in that patient's stored records.
- **Experimental risk model:** a machine-learning classifier estimates deterioration risk (low/medium/high) from structured vitals, with an explanation of contributing factors.

4.2 Core workflow

1. Nurse selects a patient and taps "Record Handoff."
2. Nurse speaks the handoff naturally; audio is recorded and uploaded.
3. Backend transcribes the audio, extracts medical entities and events, and generates an SBAR handoff, a concise summary, and timeline entries.
4. The completeness checker flags any missing required fields, and the risk model (where applicable) scores deterioration risk from structured vitals.
5. Everything is stored; the next nurse opens the patient and sees the summary, timeline, risk indicator, full handoff history, and can query the AI assistant.

4.3 Operational constraints and safety boundaries

- Keep the mobile app usable hands-free at the bedside, on standard hospital devices.
- The system provides AI-assisted organization, summarization, retrieval, and experimental risk prediction to support the handoff workflow — it does not diagnose patients or make autonomous clinical decisions. The nurse remains responsible for verifying all information.
- RAG answers must be grounded in retrieved patient records; the assistant should not generate unsupported clinical claims.
- For this student prototype, use synthetic or de-identified patient data rather than real patient recordings.

5 Solution Approach

This is a software engineering project with an applied-AI component, so the emphasis is on integrating mature, existing AI/ML components into a correct, well-tested pipeline — rather than researching new model architectures.

5.1 AI/ML component integration ...non-research

- **Speech-to-text:** use an existing pretrained model (e.g., Whisper) rather than training a speech model from scratch.
- **Entity extraction & summarization:** use an LLM/NLP pipeline via prompting or lightweight fine-tuning to pull structured fields out of transcripts and produce the SBAR/summary text.
- **RAG question answering:** chunk and embed patient records, retrieve the most relevant records for a given question, and pass only that retrieved context to the LLM so answers stay grounded in stored data instead of being generated freely.
- **Missing-information detection:** start as a rules + NLP check against an expected field checklist; consider a trained classifier later only if a suitable labeled dataset becomes available.
- **Risk classification:** compare established, well-understood models — Logistic Regression, Random Forest, XGBoost — rather than assuming one is best, and report feature-level contributions (explainability) alongside each prediction.

5.2 Mobile / backend architecture

- **Frontend:** a Flutter mobile app covering login, dashboard, patient list, patient profile, audio recorder, timeline view, and AI assistant chat.
- **Backend API:** FastAPI service exposing endpoints such as /login, /patients, /handoffs, /handoffs/{id}/process, /patients/{id}/timeline, /patients/{id}/ask, and /patients/{id}/risk.
- **Database:** PostgreSQL storing Patient, Handoff, Vitals, Medication, Timeline, and Risk tables with clear relations back to each patient.

5.3 CI/CD and fast delivery enabler

- Automatically build and run backend unit tests and linting on every change.
- Produce repeatable deployment artifacts to a staging environment.

- Enable safe, frequent releases of each incremental stage in the roadmap (backend → database → patient system → mobile app → audio → speech AI → NLP → summary/timeline → RAG → ML → testing → deployment).

6 Evaluation Criterion

We define success using measurable metrics tied to the core workflow.

6.1 Primary evaluation metric

Handoff Documentation Time (HDT): time from the start of voice recording to a fully structured, stored handoff record (transcript + SBAR + summary + timeline entries).

- Target: median HDT well below the time required to manually type an equivalent handoff, measured in a small user study.
- Attribution: measured from timestamps logged at recording start and at the handoff-stored event in the database.

6.2 Secondary evaluation metrics

- **Entity extraction accuracy:** precision/recall/F1 of extracted fields (diagnosis, medication, vitals, allergy, tests) against manually labeled transcripts.
- **Completeness-detection accuracy:** precision/recall of the missing-information checker against handoffs with intentionally omitted fields.
- **RAG answer groundedness:** proportion of AI assistant answers that are fully supported by the retrieved patient records, reviewed manually on a test set of questions.
- **Risk model performance:** accuracy, precision, recall, F1, and ROC-AUC, with particular attention to recall/sensitivity and false negatives given the medical stakes.
- **Nurse-reported usability:** clarity and usefulness of summaries/timeline using a simple 1–5 feedback question.
- **Reliability:** availability of the staging/production API during testing sessions.

6.3 Pilot validation plan ...high-level

- Recruit a small group of nursing students or clinically-informed volunteers (e.g., 5–10) to record handoffs for synthetic patient scenarios.
- Compare structured-handoff creation time against a manual-typing baseline for the same scenarios.

- Collect qualitative feedback on summary/timeline usefulness, missing-information flags, and AI assistant answer quality.

7 Scalability

1. **Scalable (theoretically):** the system should support growth in number of patients, handoffs, and concurrent app usage by:
    - decoupling the Flutter frontend from the FastAPI backend,
    - using a stateless REST API design,
    - indexing and paginating common queries (patient lists, timelines),
    - processing audio transcription asynchronously (queue-based) so long-running AI jobs don't block the API.
2. **Operationally deployable (practically):** the system should run on common infrastructure:
    - a managed PostgreSQL database,
    - containerized backend deployment,
    - a CI/CD pipeline for staging/production,
    - a vector store for RAG retrieval that can scale independently of the relational database.

8 Engine availability heuristic

A mature set of ”engines” (tooling/services) is available to implement this project without inventing new infrastructure or model architectures:

- Speech-to-text engine (e.g., Whisper) for audio $\rightarrow$ transcript.
- Hosted or open-source LLM for entity extraction, SBAR structuring, summarization, and RAG-based answer generation.
- Embedding + vector search library for retrieval-augmented generation.
- FastAPI as the REST framework; Flutter as the cross-platform mobile framework.
- Managed PostgreSQL for relational storage.
- Token-based authentication (e.g., JWT).
- Object storage for audio files.
- scikit-learn / XGBoost for risk classification experiments.

- pytest for backend unit/integration tests; a CI runner and staging deployment target.

We will use these mature components to focus on pipeline design, data structuring, correctness, and clinical-safety framing rather than building new infrastructure or novel model architectures.

9 Project scope and deliverables

9.1 Initial deliverable ...first iteration

- Login/authentication and a Flutter app shell connected to the FastAPI backend.
- Patient list and patient profile screens backed by a PostgreSQL schema (Patient, Handoff, Vitals, Medication, Timeline, Risk).
- Manual handoff entry and retrieval (structured fields, no AI yet), to validate the core data model end to end.
- CI: build + backend unit tests + linting.
- CD to staging.

9.2 Subsequent deliverables

- Audio recording, upload, and storage.
- Whisper-based speech-to-text integration.
- Medical NLP entity extraction feeding an auto-generated SBAR handoff and AI summary.
- Automatic timeline generation from extracted events.
- Missing-information / completeness checker.
- RAG-based AI assistant for patient-specific question answering.
- Experimental ML deterioration risk model with explainability, plus expanded testing and deployment.

10 Risks and mitigations

- **Patient data privacy/consent:** use only synthetic or de-identified data for the prototype; no real patient recordings; plan for stronger safeguards (access control, audit logs) before any real deployment.

- **Overclaiming AI capability:** explicitly label the risk model and summaries as decision-support, not diagnosis; keep the nurse responsible for verification; constrain RAG answers to retrieved records to avoid hallucination.
- **Speech recognition accuracy in clinical/noisy settings:** evaluate transcription quality on medical vocabulary; give the nurse a quick review/correction step before a handoff is finalized.
- **False negatives in risk prediction:** prioritize recall/sensitivity over raw accuracy; tune thresholds; keep a human in the loop for any high-risk flag.
- **Staff adoption:** keep the voice-recording workflow faster than typing; make structured output easy to review and edit; minimize extra steps versus current practice.
- **Scope creep across many AI/ML stages:** follow the staged roadmap and ship each stage (backend → database → frontend → audio → speech AI → NLP → summary/timeline → RAG → ML → testing → deployment) as a working increment before starting the next.

11 Summary

NurseSync AI proposes a deployable, voice-driven clinical handoff system that converts spoken nurse-to-nurse handoffs into structured, timestamped, and searchable clinical records, augmented with AI summarization, automatic timeline generation, missing-information detection, and a RAG-grounded assistant for patient-specific question answering. An experimental, explainable ML deterioration-risk model adds a decision-support layer without replacing clinical judgment. The project emphasizes engineering integration of mature, existing AI/ML components, incremental delivery through CI/CD, and measurable evaluation — handoff documentation time, extraction accuracy, and risk-model recall — while maintaining clear safety boundaries: the system organizes, summarizes, and retrieves information to support the nurse, and does not diagnose or make autonomous care decisions.